

Chapter 7: Portfolio Construction & NCO

Machine Learning for Asset Managers (Marcos M. López de Prado)

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Chapter 7 Overview: The Failure of Convex Optimization

The Promise of Markowitz (1952)

- Formulates portfolio allocation as a quadratic programming problem (Critical Line Algorithm – CLA).
- Identifies the **Efficient Frontier**: portfolios maximizing expected return for a given level of risk.
- Mathematically sound and foundational to modern portfolio theory.

The Out-of-Sample Reality

- **Extreme Instability**: Small deviations in estimated mean μ or covariance V produce wildly erratic allocations.
- **Extreme Concentration**: Often discards most assets in favor of heavy bets on a few holdings.
- **Underperformance**: Naïve $1/N$ equal-weighting frequently beats Markowitz out-of-sample (DeMiguel et al., 2009).

Topic 7.1: Convex Portfolio Optimization Framework

Quadratic Programming Formulation

For N assets with excess expected returns μ and covariance V , Markowitz solves:

$$\min_{\omega} \frac{1}{2} \omega' V \omega \quad \text{subject to } \omega' a = 1$$

where a represents the constraint vector. The closed-form analytical solution is:

$$\omega^* = \frac{V^{-1} a}{a' V^{-1} a}$$

Special Cases of the Characteristic Vector a

- **Equal Weights ($1/N$):** $a = 1_N$ and $V = \sigma I_N \implies \omega^* = \frac{1}{N} 1_N$.
- **Inverse Variance (IVP):** $a = 1_N$ and $V = \text{diag}(V_{i,i}) \implies \omega_i^* \propto \frac{1}{V_{i,i}}$ (ignores correlations).
- **Minimum Variance Portfolio:** $a = 1_N \implies \omega^* = \frac{V^{-1} 1_N}{1_N' V^{-1} 1_N}$ (depends purely on V).
- **Maximum Sharpe Ratio Portfolio:** $a = \mu \implies \omega^* = \frac{V^{-1} \mu}{1_N' V^{-1} \mu}$ (highly sensitive to μ and V).

Topic 7.2: The Condition Number & Matrix Inversion Explosion

Eigenvalues of 2×2 Correlation Matrix

Consider two assets with correlation $\rho \in (-1, 1)$:

$$C = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

Eigenvalues are the roots of $|C - \lambda I| = 0$:

$$\lambda_1 = 1 + \rho, \quad \lambda_2 = 1 - \rho$$

- **Determinant:** $|C| = \lambda_1 \lambda_2 = 1 - \rho^2$.
- Maximized at $\rho = 0 \implies \lambda_1 = \lambda_2 = 1$.
- Approaches 0 as $|\rho| \rightarrow 1$.

Condition Number (κ)

$$\kappa(C) = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{1 + |\rho|}{1 - |\rho|}$$

The Inversion Explosion:

$$C^{-1} = \frac{1}{1 - \rho^2} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$$

As $|\rho| \rightarrow 1$, $\kappa(C) \rightarrow \infty$ and elements of C^{-1} explode. Estimation noise is magnified exponentially during inversion!

Topic 7.3: Markowitz's Curse (The Correlation Paradox)

The Paradox of Classical Optimization

Markowitz's Curse $\implies$ The more correlated the investments, the more unstable the solution!

Why We Need Optimization

- When assets are highly correlated ($\rho \neq 0$), portfolio managers desperately need optimization to construct diversified baskets.
- When assets are uncorrelated ($\rho \approx 0$), V is diagonal, and simple Inverse Variance (IVP) is already optimal.

Why Optimization Fails

- High correlation $\rho \rightarrow 1$ causes high condition numbers $\kappa(V) \rightarrow \infty$.
- V^{-1} becomes numerically ill-conditioned and ultra-sensitive to small noise.
- **Conclusion:** Markowitz is numerically stable only when $\rho \approx 0$ – precisely when we do NOT need it!

Topic 7.4: Signal vs. Noise as Sources of Instability

1. Noise-Induced Instability (Low T/N Ratio)

- Driven by finite sample size relative to dimension (N/T).
- Governed by the Marcenko-Pastur distribution (Chapter 2).
- As $N/T \rightarrow 1$, sample noise creates artificial near-zero eigenvalues.
- **Remedy:** Denoising via Random Matrix Theory (RMT) or shrinkage.

2. Signal-Induced Instability (Data Structure)

- Driven by true economic correlation clusters (sectors, factors).
- **Independent of sample size T :** sampling more data does NOT remove it!
- One dominant eigenvector in a cluster absorbs variance. Since $\text{tr}(C) = N$, other $K - 1$ eigenvalues shrink to ≈ 0 .
- **Remedy: Nested Clustered Optimization (NCO).**

Topic 7.5: The Nested Clustered Optimization (NCO) Framework

What is NCO?

NCO is a modular **wrapper algorithm** designed by Marcos López de Prado to eliminate Markowitz's Curse by containing numerical instability inside correlation clusters.

1. Correlation Clustering

- Denoise V using RMT (Chapter 2).
- Partition assets into K clusters $C_1, \dots, C_K$ using ONC (Chapter 4).

2. Intracluster Allocations

- Solve optimal weights ω_k^{intra} independently within each cluster C_k .
- Contains high condition numbers inside sub-matrices!

3. Intercluster Allocations

- Reduce V to intercluster covariance
 $V_{\text{inter}} = \omega'_{\text{intra}} V \omega_{\text{intra}}$.
- V_{inter} is nearly diagonal ($\rho \approx 0$). Solve ω^{inter} .

Final NCO Allocation

$$\omega_{\text{NCO}} = \omega_{\text{intra}} \odot \omega_{\text{inter}}$$

Topic 7.6: Step-by-Step NCO Mathematical Mechanics

Step 1: Intracluster Optimization

For each cluster $k \in \{1, \dots, K\}$ spanning asset indices C_k :

$$\min_{\omega_k} \frac{1}{2} \omega_k' V_{C_k, C_k} \omega_k \quad \text{s.t.} \quad \omega_k' a_{C_k} = 1 \implies \omega_k^{\text{intra}}$$

This forms an $N \times K$ intracluster weight matrix W_{intra} .

Step 2: Intercluster Covariance Reduction

Compute the reduced $K \times K$ intercluster covariance matrix representing cluster returns:

$$V_{\text{inter}} = W_{\text{intra}}' V W_{\text{intra}}, \quad \mu_{\text{inter}} = W_{\text{intra}}' \mu$$

Because intercluster correlations are low, V_{inter} has a condition number near 1 (well-behaved Markowitz case!).

Step 3: Intercluster Optimization & Final Weights

$$\min \frac{1}{2} \omega_{\text{inter}}' V_{\text{inter}} \omega_{\text{inter}} \quad \text{s.t.} \quad \omega_{\text{inter}}' a_{\text{inter}} = 1 \implies \omega_{\text{inter}}^*$$

Topic 7.7: NCO vs. Hierarchical Risk Parity (HRP) vs. Markowitz

Feature	Markowitz (CLA)	Inverse Variance (IVP)	NCO / HRP (ML Methods)
Matrix Inversion?	Required (V^{-1})	None (Diagonal only)	None (HRP) / Segmented (NCO)
Condition Number Risk?	High (Explodes)	Zero (Ignores correlation)	Low (Contained in clusters)
Asset Structure?	Fully-connected graph	Unconnected nodes	Tree / Hierarchical clusters
Weight Concentration?	Extreme (Heavy bets)	Uniform (Ignores risk)	Balanced across clusters
OOS Performance?	High variance / fragile	Robust but suboptimal	Optimal & Lowest OOS Variance

Key Takeaway

Markowitz evaluates all $N(N-1)/2$ graph edges simultaneously, allowing noise to distort every weight. ML methods (NCO and HRP) impose tree-like / cluster structures, restricting weight rebalancing to true peers.

Topic 7.8: Empirical Results – Monte Carlo Experiments

Minimum Variance Portfolio (50 Assets)

Monte Carlo simulation (1,000 trials, 10 block clusters with $\rho_{\text{intra}} = 0.5$):

- **Markowitz Raw RMSE:** 7.95×10^{-3}
- **Ledoit-Wolf Shrinkage RMSE:** 8.89×10^{-3}
- **NCO RMSE:** 4.21×10^{-3}
- **RMSE Reduction:** 47.02% error reduction from NCO alone! Shrinkage adds no extra value.

Maximum Sharpe Ratio Portfolio

Monte Carlo simulation evaluating optimal tangency allocations:

- **Markowitz Raw RMSE:** 7.02×10^{-2}
- **Ledoit-Wolf Shrinkage RMSE:** 6.54×10^{-2}
- **NCO RMSE:** 3.17×10^{-2}
- **RMSE Reduction:** 54.83% error reduction by replacing Markowitz with NCO!

Summary: Key Takeaways on Portfolio Construction

Core Principles for Quantitative Asset Managers

- 1 **Recognize Markowitz's Curse:** Quadratic optimization requires V^{-1} , which explodes under high correlation. The more you need diversification, the less reliable classical optimization becomes.
- 2 **Distinguish Signal from Noise:** Sampling more data (T) fixes noise-induced instability, but does NOT fix structural signal-induced instability caused by correlation clusters.
- 3 **Adopt NCO Wrapper Architecture:** NCO splits portfolio allocation into intracluster and intercluster sub-problems, containing condition numbers inside sub-matrices and converting V_{inter} into a well-behaved diagonal matrix.
- 4 **Modular Flexibility:** NCO is agnostic to the objective function – it works seamlessly with Minimum Variance, Maximum Sharpe Ratio, Black-Litterman, or custom PM constraints.

Thank You!

Questions & Discussion

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